

Prove the monotone convergence theorem:

Theorem: Given a sequence $\{f_n\}$ of non-negative measurable functions $f_n : X \rightarrow \mathbb{R}$, such that $f_n \leq f_{n+1}$. Let $f = \lim_{n \rightarrow \infty} f_n$, then

$$\lim_{n \rightarrow \infty} \int_X f_n \, d\mu = \int_X f \, d\mu.$$

Proof: Clearly $\int_X f_n \, d\mu \leq \int_X f \, d\mu$, we show that for any non-negative simple function $0 \leq g \leq f$ and $\varepsilon > 0$, $\lim_{n \rightarrow \infty} \int_X f_n \, d\mu \geq (1 - \varepsilon) \int_X g \, d\mu$.

Let $A_n = \{x \in X : f_n(x) \geq (1 - \varepsilon)g(x)\}$, then $A_n \subset A_{n+1}$ and $X = \bigcup_{n \geq 1} A_n$. Hence

$$\lim_{n \rightarrow \infty} \int_X f_n \, d\mu \geq \lim_{n \rightarrow \infty} \int_{A_n} f_n \, d\mu \geq \lim_{n \rightarrow \infty} \int_{A_n} (1 - \varepsilon)g \, d\mu = \lim_{n \rightarrow \infty} \mu_{(1-\varepsilon)g}(A_n) = \mu_{(1-\varepsilon)g}(X).$$

Where $\mu_{(1-\varepsilon)g}(A) = \int_A (1 - \varepsilon)g \, d\mu$ is a measure on X . $\square$